Assignment 2 | Final Testing Report

Student Course Registration System | Black-Box + White-Box Testing | Defect Management | GitHub Collaboration Preparation

DELIVERABLE	VERIFIED RESULT
Black-box tests	62 tests; original instructor starter: 44 passed / 18 failed
White-box additions	18 tests; structural tests documented WB001-WB018
Pre-fix combined run	53 passed / 27 failed; original defects retained for evidence
Corrected application	80 passed / 0 failed
Statement / branch coverage	96.49% / 92.31% initially; 100% / 100% after verified corrections
Confirmed distinct defects	5 defects identified and locally corrected; real GitHub Issues/PRs require account and teammates

Testing source of truth: REQUIREMENTS.md (FR-01 through FR-15); original starter source and test logs preserved in docs/evidence. Report generated from actual local pytest and pytest-cov runs.

1. Scope, requirements, and test process

The provided registration.py was tested without replacing it with a new system. Test oracles were derived from the SRS first. The original tests were executed and legitimate failures preserved; only then were structural tests designed and five minimal fixes applied. All test expectations remained unchanged.

- Input constraints: current credits 0-18, course credits 1-4, dues >=0, capacity >=1 and 0 <= enrolled <= capacity.
- Decision priority (after input validation): dues, already registered, prerequisites, credit limit, then seat availability.
- Registration must update both student and course state; waitlisting must not affect credits or enrollment; withdrawal must decrement enrollment.

2. Part A - Black-box test design

A1. Equivalence class partitioning (22 design rows)

Both valid and invalid classes were identified for dues, prerequisites, registration state, combined credits, course credits, and course availability. Each row records requirement, class, representative input, expected result and canonical BB test ID in docs/blackbox_test_design.xlsx (ECP tab).

EXAMPLE CLASS	REPRESENTATIVE	EXPECTED	TEST
Zero dues	dues=0	REGISTERED	BB001
Any positive dues	dues=1	DUES_BLOCK	BB002
Negative dues	dues=-1	ValueError: Invalid dues	BB027
Partial prerequisites	completed={CS101}; required={CS101,MATH101}	PREREQUISITES_MISSING	BB006
All prerequisites	completed={CS101,MATH101}; required equal	REGISTERED	BB008
Exactly 18 credits	15+3	REGISTERED	BB010
Course already full	10 enrolled / 10 capacity	WAITLISTED	BB011
Course credits above allowed	new credits=5	ValueError: Invalid course credits	BB021

A2. Boundary value analysis (26 design rows)

Boundaries include current credits -1, 0, 17, 18, 19; course credits 0, 1, 4, 5; combined 17/18/19; enrollment -1/0/capacity-1/capacity/capacity+1; capacity 0/1; dues -1/0/1. Each expected output is linked to an executable canonical test ID.

BOUNDARY	TEST	EXPECTED
Current credits -1 / 0 / 19	BB018 / BB013 / BB019	error / registered / error
Course credits 0 / 1 / 4 / 5	BB020 / BB013 / BB016 / BB021	error / valid / valid / error
Combined credits 17 / 18 / 19	BB015 / BB010 / BB009	registered / registered / credit limit
Seat count capacity-1 / capacity / capacity+1	BB012 / BB011 / BB026	registered / waitlisted / error
Dues -1 / 0 / 1	BB027 / BB001 / BB002	error / registered / dues block

A3. Decision-table test design (14 rules)

Assume numeric inputs valid for the five-rule decision table. Y means a condition holds, N means not; first applicable outcome wins. Validation precedence cases BB033-BB036 are listed separately in the decision_table.xlsx workbook.

RULE	D+	DUP	PRE	CRED	FULL	FIRST RESULT	TEST
D01	Y	Y	Y	Y	Y	DUES_BLOCK	BB028
D02	N	Y	Y	Y	Y	ALREADY_REGISTERED	BB029
D03	N	N	Y	Y	Y	PREREQUISITES_MISSING	BB030
D04	N	N	N	Y	Y	CREDIT_LIMIT	BB031
D05	N	N	N	N	Y	WAITLISTED	BB011
D06	N	N	N	N	N	REGISTERED	BB001
D07	Y	N	N	N	N	DUES_BLOCK	BB002
D08	N	Y	N	N	N	ALREADY_REGISTERED	BB005
D09	N	N	Y	N	N	PREREQUISITES_MISSING	BB006
D10	N	N	N	Y	N	CREDIT_LIMIT	BB009
D11	N	N	N	N	Y	WAITLISTED	BB032
D12	N	N	N	N	N	REGISTERED	BB008
D13	N	N	N	N	Y	WAITLISTED	BB024
D14	Y	N	N	N	N	DUES_BLOCK	BB003

A4/A5. Executable tests and exact-duplicate optimization

The initial black-box run executed 62 canonical tests, yielding 44 passes and 18 failures on the unmodified starter. For state updates BB052-BB062, pytest assertions verify registered courses, credits, enrollment and waitlist; unsuccessful actions must leave the pre-call state unchanged. Validation errors use pytest.raises() with exact messages.

Optimization method: same function + exact same inputs + exact same expected output = a genuine duplicate. The design sheets reference shared canonical IDs instead of executing duplicate copies. There are 19 canonical cases referenced by multiple ECP/BVA/decision rows, with cross-technique traceability in the Optimization tab. Matching outcomes alone were NOT treated as duplication.

3. Part B - white-box structural analysis

B1. CFG and cyclomatic complexity

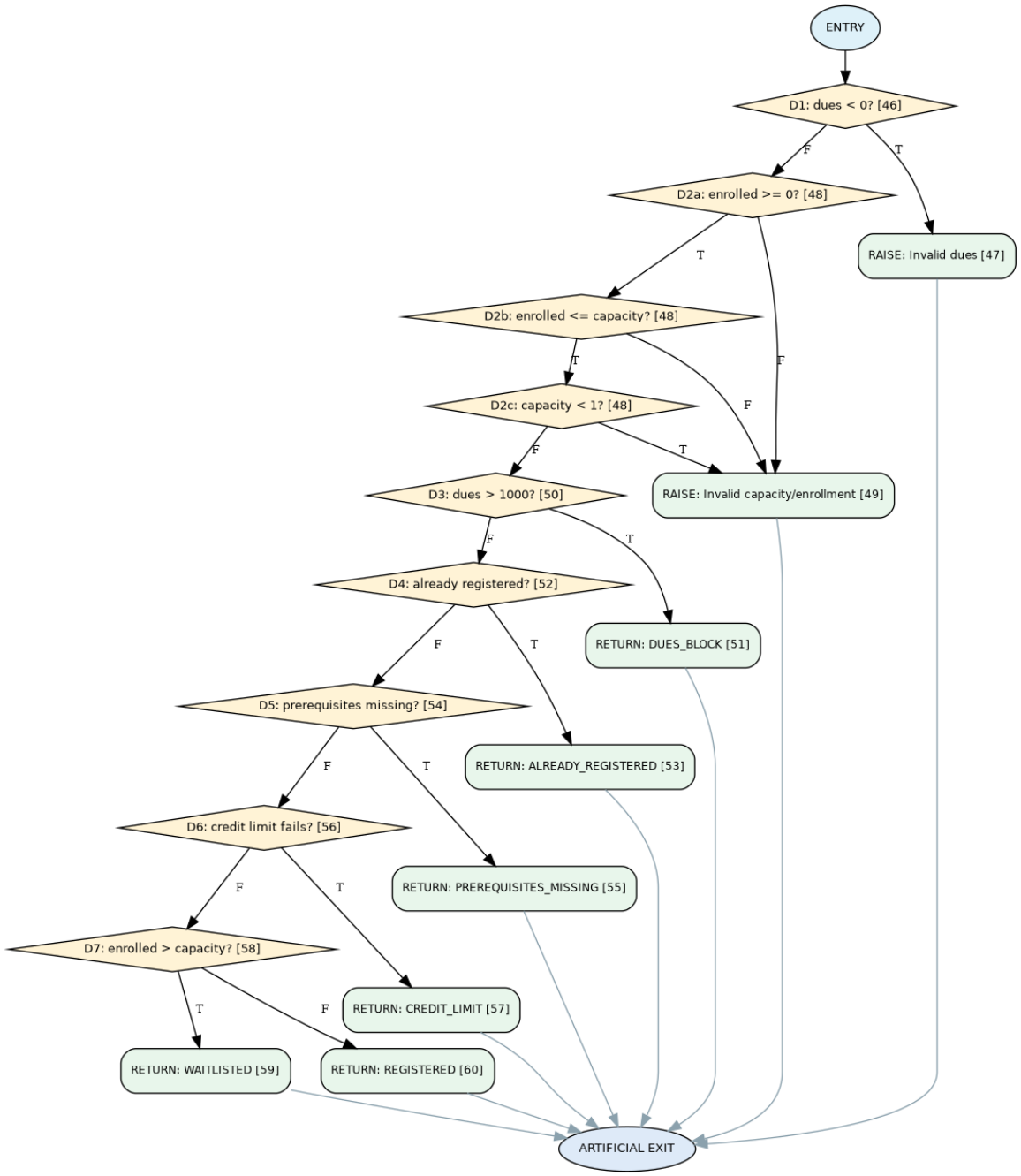


Figure 1. Original registration_decision() CFG. Diamonds are decisions; rounded boxes are return/exception endpoints. Synthetic EXIT connects terminal nodes for correct E-N+2 accounting. D2a/b/c show Python short-circuit evaluation explicitly.

Expanded short-circuit convention: count 0 <= enrolled, enrolled <= capacity, and capacity < 1 separately. The original source has 9 decision nodes (D1, D2a/b/c, D3-D7), 8 terminal nodes, 1 ENTRY and 1 synthetic EXIT: N = 19; E = 27; V(G) = E - N + 2 = 27 - 19 + 2 = 10. The decision-node cross-check is 9 + 1 = 10.

Collapsed convention: treat line 48 as one compound decision. Then N = 17; E = 23; V(G) = 23 - 17 + 2 = 8, equal to 7 compound decision statements + 1. All other compound and helper calls are treated as a single predicate node at the call site. The graph depicts the expanded convention.

Independent basis paths (syntactic basis; 10 paths)

PATH	DISTINCT BRANCH	TERMINAL OUTCOME	REP TEST
P1	D1=T	Invalid dues	BB027
P2	D1=F, D2a=F	Invalid enrollment below 0	BB025
P3	D2a=T, D2b=F	Invalid enrollment above capacity	BB026
P4	D2a=T, D2b=T, D2c=T	Invalid capacity 0	BB022
P5	D2c=F, D3=T	DUES_BLOCK	BB004
P6	D3=F, D4=T	ALREADY_REGISTERED	BB005
P7	D4=F, D5=T	PREREQUISITES_MISSING	BB007
P8	D5=F, D6=T	CREDIT_LIMIT	BB014
P9	D6=F, D7=T	WAITLISTED: infeasible in original	BB011 / WB014 (fix only)
P10	D7=F	REGISTERED	BB001

Important feasibility constraint: the original D7 condition is enrolled > capacity, but D2b would reject that input earlier. Thus P9 is a syntactic basis path that CANNOT be exercised in the unmodified code; this explains an uncovered branch and is a confirmed defect, not missing test effort. After correction D7 uses enrolled == capacity and P9 becomes feasible.

B2. Initial vs. extended vs. corrected coverage

RUN	PASS / FAIL	STATEMENT	BRANCH	MISSING LINES
Original: BB only	44 / 18	55/57 (96.49%)	24/26 (92.31%)	59, 71
Original: BB + WB	53 / 27	55/57 (96.49%)	24/26 (92.31%)	59, 71
Corrected: BB + WB	80 / 0	58/58 (100%)	26/26 (100%)	None

Original missing branch arcs were (58 -> 59) and (70 -> 71). Because registration_decision() never produced WAITLISTED, both the waiting-list append and its decision branch were unreachable. White-box additions targeted input checks, early exits, exception ordering and short-circuit paths (WB001-WB018); they could not force logically unreachable branches. The defect-fix branch made them reachable and the unchanged expectations all passed.

B3. Atomic condition and short-circuit truth tables

CAPACITY VALIDATION CASE	A: enrolled>=0	B: enrolled<=capacity	C: capacity<1	RAISE?	TEST
enrolled=0; cap=10	T	T	F	F	BB001
enrolled=-1; cap=10	F	NE	NE	T	BB025
enrolled=11; cap=10	T	F	NE	T	BB026
enrolled=0; cap=0	T	T	T	T	BB022
enrolled=0; cap=-1	T	F	NE	T	BB036

A = 0 <= enrolled; B = enrolled <= capacity; C = capacity < 1. Actual expression is not (A and B) or C. NE = not evaluated because Python uses short-circuit operators and comparison chaining. The table is manual condition analysis, not a pytest-cov MC/DC result.

ATOMIC GUARD	TRUE EXAMPLE	FALSE EXAMPLE	NE EXAMPLE
D1: dues < 0	BB027	BB001	-

ATOMIC GUARD	TRUE EXAMPLE	FALSE EXAMPLE	NE EXAMPLE
D2a: enrolled >= 0	BB001	BB025	BB027
D2b: enrolled <= capacity	BB001	BB026	BB025
D2c: capacity < 1	BB022	BB001	BB026
D3: dues > starter threshold	BB004	BB001	BB027
D4: duplicate	BB005	BB001	BB004
D5: prerequisites missing	BB007	BB001	BB005
D6: credits fail	BB014	BB001	BB007
D7: original seats > capacity	infeasible	BB001	BB014

All original syntactically reachable decision outcomes are represented. The original D7 true condition is infeasible, so full condition coverage on the original program is impossible. After correction, D7 true is exercised by BB011/WB014 and false by BB001.

MC/DC - two-condition waitlist append in register_course()

Chosen compound expression (the elif guard): C1 = decision == WAITLISTED; C2 = student ID not already in waitlist. The boolean outcome controls whether to append the ID; it is reached only when first REGISTERED branch is false. For a C1=False case we therefore use a dues-blocked decision (not REGISTERED).

TEST	C1	C2	COMPOUND	EFFECT
WB009	T	T	T	ID appended
WB010	F	NE	F	DUES_BLOCK, no append
WB011	T	F	F	Already waitlisted, no append
WB012	T	T	T	ID appended

Independence pairs: C1: WB009 vs WB010 (the second operand is not evaluated when C1 is false; logically the waitlist is empty in both). C2: WB011 vs WB012 (C1 remains true and only membership changes). Two of two independently variable conditions demonstrated, therefore 2/2 = 100% manual MC/DC for this SELECTED compound decision. This is not a claim that the entire codebase has 100% MC/DC.

B4. White-box test purposes

WB001-WB004: input validation and early exits. WB005-WB006: empty and partially missing prerequisites. WB007-WB008: lower bound of credit validation. WB009-WB012: short-circuit waitlist MC/DC. WB013-WB014: exact-credit and full-capacity branches. WB015-WB016: withdrawal success/failure. WB017: due-versus-duplicate priority. WB018: invalid course credit input. Test IDs are in the source function names/comments.

4. Part C - defects and GitHub collaboration

ISSUE DRAFT	REQ	ORIGINAL DEFECT	FAILING IDs	LOCAL STATUS
DEF-01	FR-01/07	Exactly 18 credits wrongly rejected	BB010, BB039	Fixed locally
DEF-02	FR-03/06	Partial prerequisites wrongly accepted	BB006, BB049	Fixed locally
DEF-03	FR-04/09	Dues Rs. 1 ignored	BB002, BB055	Fixed locally
DEF-04	FR-08/12	Course at capacity not waitlisted	BB011, BB053	Fixed locally
DEF-05	FR-13	Withdrawal enrollment not decremented	BB059, BB061	Fixed locally

Each docs/github_issue_drafts/DEF-0x.md records reproducible steps, expected/actual outputs, requirement IDs, failed tests and corrective action. These are issue-ready drafts ONLY: no real issue URL has been created. Defect fixes were made in the separate local defect-fixes branch; the original starter is preserved in git main and as evidence.

Local git history was prepared with main, blackbox-testing, whitebox-testing and defect-fixes branches; descriptive commits and a pytest/coverage GitHub Actions workflow are included. The workflow uploads test and coverage artifacts even when tests fail, preserving legitimate failing runs after branches are pushed to GitHub.

GitHub repository URL: NOT YET CREATED. Issue URLs: NOT YET CREATED. PR URLs: NOT YET CREATED. GitHub Actions run URLs: NOT YET CREATED. Peer reviews: require actual teammates. The exact steps and PR descriptions are supplied in SUBMISSION_GUIDE.md and docs/PULL_REQUEST_DRAFTS.md. Do not submit local files as fictitious online collaboration evidence.

5. Part D - conclusion and submission references

The SRS-driven suite exposed five independent functional defects on the supplied system. The test oracle was not changed to make the defective version pass. Structural analysis showed an infeasible waitlist path, explaining why a high initial statement coverage percentage did not imply correctness. After correcting the five defects, the original requirements-based tests and extra structural tests passed with measured 100% statement and branch coverage.

- Source and tests: registration.py; tests/test_blackbox.py; tests/test_whitebox.py; REQUIREMENTS.md.
- Black-box design: docs/blackbox_test_design.xlsx and docs/decision_table.xlsx.
- Structural analysis: docs/cfg.png, docs/cfg_fixed.png and this report.
- Verified evidence: docs/evidence/initial_blackbox_pytest.txt, pre_fix_all_pytest.txt and final_pytest.txt; accompanying coverage JSON/XML.
- GitHub deliverables: .github/workflows/tests.yml, docs/github_issue_drafts, docs/PULL_REQUEST_DRAFTS.md and SUBMISSION_GUIDE.md.
- Before submission, insert real group member contributions, repository address, actual Issues, PRs and reviewer links from your own GitHub account.

Document limitation: local testing and local Git activity were verified; cloud GitHub actions and human peer reviews cannot be truthfully generated without access to the required GitHub account and collaborators.